

7 Fill your pitcher with water and add in as much of the soil as you like. Stir it all up with a spoon, so that the soil mixes in thoroughly. The smallest pieces of soil will be “suspended”—left floating—in the water, and some parts will dissolve.

Soil particles contain tiny living organisms, such as bacteria.

These objects from nature won't look pretty for long, as they get mixed up in the dirty water.



8 Drop in a few leaves and blades of grass. Your water is now well and truly dirty! It contains particles of many different sizes and various dissolved substances. Any of these things might make you ill if you drank the water.

In a river or pond, leaves and grass would float on top of the water.



Don't forget to wash the pitcher thoroughly when you've finished your experiment!

9 Slowly pour some dirty water onto the pebbles at the top of your filter. Hold the filter steady to make sure it doesn't topple over. Watch as the water trickles through the layers, and emerges much cleaner at the bottom.